Antiderivatives and Indefinite Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find each indefinite integral:

a $\int (3x^2 - 4x + 5)\,dx = x^3 - 2x^2 + 5x + C$

b $\int \frac{1}{x^2}\,dx = -\frac{1}{x} + C$

c $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C$

d $\int \sec^2 x\,dx = \tan x + C$

(a) $= x^3 - 2x^2 + 5x + C$; (b) $= -\frac{1}{x} + C$; (c) $= \frac{2}{3}x^{3/2} + C$; (d) $= \tan x + C$

2 [Foundational] • Target: Using initial conditions

A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

$s(t) = \int v(t)\,dt = t^3 - 3t^2 + C$. Using $s(0) = 4$: $C = 4$: $s(t) = t^3 - 3t^2 + 4$.

3 [Foundational] • Target: Using initial conditions

An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s^2.

a Find the velocity function $v(t)$. $v(t) = -9.8t$ m/s (using $v(0) = 0$).

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$. $s(t) = -4.9t^2 + 45$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places. $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

(a) $v(t) = -9.8t$ m/s (using $v(0) = 0$).; (b) $s(t) = -4.9t^2 + 45$.; (c) $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

4 [Foundational] • Target: More antiderivatives

Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

Treating $(x + 1)$ as a single unit (power rule for a linear inner function): $\int (x + 1)^3\,dx = \frac{(x+1)^4}{4} + C$.

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5 [Foundational] • Target: Motion applications

A rocket's acceleration is $a(t) = 6t$ m/s^2, with $v(0) = 0$ and $s(0) = 0$.

a Find $v(t)$. $v(t) = 3t^2$.

b Find $s(t)$. $s(t) = t^3$.

c Find the distance traveled in the first 4 seconds. $s(4) = 64$ m.

(a) $v(t) = 3t^2$.; (b) $s(t) = t^3$.; (c) $s(4) = 64$ m.

6 [Foundational] • Target: Antiderivatives of composite expressions

Find $\int (2x - 1)^5 \, dx$, treating $(2x - 1)$ as a linear inner function.

$\int (2x-1)^5 \, dx = \frac{(2x-1)^6}{6 \cdot 2} + C = \frac{(2x-1)^6}{12} + C$ (dividing by the derivative of the inner function, 2).

7 [Foundational] • Target: Verifying antiderivatives by differentiation

Verify that $F(x) = x\ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.

$F'(x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x = f(x)$ ✓.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: A projectile motion application

A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8$ m/s^2. Find its velocity function $v(t)$ and position function $s(t)$.

$v(t) = \int a(t) \, dt = -9.8t + C_1$. Using $v(0) = 15$: $v(t) = -9.8t + 15$.
$s(t) = \int v(t) \, dt = -4.9t^2 + 15t + C_2$. Using $s(0) = 2$: $s(t) = -4.9t^2 + 15t + 2$.

9 [Intermediate] • Target: Visualizing an antiderivative family

The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F.

a Write the general antiderivative of $f(x) = 2x$. $F(x) = x^2 + C$.

b If a specific antiderivative passes through $(1, 4)$, find its constant C. $1^2 + C = 4$, so $C = 3$: $F(x) = x^2 + 3$.

(a) $F(x) = x^2 + C$.; (b) $1^2 + C = 4$, so $C = 3$: $F(x) = x^2 + 3$.

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10 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int \frac{x^3+2x}{x}\,dx$ by simplifying the integrand before integrating.

$\frac{x^3+2x}{x} = x^2+2$, so $\int (x^2+2)\,dx = \frac{x^3}{3}+2x+C$.

11 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int (x+1)(x-3)\,dx$ by expanding the product first.

$(x+1)(x-3) = x^2-2x-3$, so $\int (x^2-2x-3)\,dx = \frac{x^3}{3}-x^2-3x+C$.

12 [Intermediate] • Target: Using initial conditions

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x+1)\ln 3 = (x-2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

13 [Intermediate] • Target: More antiderivatives

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0) = 2$.

Identify $P(x) = -1$ and $Q(x) = e^{3x}$. Compute integrating factor: $\mu(x) = e^{\int -1\,dx} = e^{-x}$. Multiply through: $\frac{d}{dx}[e^{-x}y] = e^{-x}e^{3x} = e^{2x}$ Integrate both sides: $e^{-x}y = \int e^{2x}\,dx = \frac{1}{2}e^{2x} + C \Rightarrow y(x) = \frac{1}{2}e^{3x} + Ce^{x}$ Apply initial condition $y(0) = 2$: $2 = \frac{1}{2}(1) + C(1) \Rightarrow C = \frac{3}{2}$ Thus, $y(x) = \frac{1}{2}e^{3x} + \frac{3}{2}e^{x}$.

14 [Intermediate] • Target: Motion applications

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

Find intersection points: $x^2 = 2x - x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow 2x(x-1) = 0$, so $x = 0$ and $x = 1$. On $[0, 1]$, test $x = 0.5$: $2(0.5) - (0.5)^2 = 0.75 > (0.5)^2 = 0.25$, so $y = 2x - x^2$ is the upper curve.

$A = \int_0^1 [(2x - x^2) - x^2]\,dx = \int_0^1 (2x - 2x^2)\,dx = \left[x^2 - \frac{2x^3}{3}\right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

Substitute $u = \ln x$. The derivative is $du = \frac{1}{x}\,dx$. The integral simplifies directly to: $\int \cos(u)\,du = \sin(u) + C = \sin(\ln x) + C$

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16 **[Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions**

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 **[Challenging] • Target: Antiderivatives of reciprocal and rational-power functions • Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 **[Challenging] • Target: A projectile motion application • Cross-Lesson: 15.01 Double Integrals**

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging] • Target: Visualizing an antiderivative family • Cross-Lesson: 16.01 Vector Fields and Line Integrals**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging] • Target: Antiderivatives requiring algebraic rewriting first • Cross-Lesson: 2.01 Evaluating Limits**

Level 3 · AdvancedLet $f(x) = x^5 + 3x^3 + 2x - 1$. Find the equation of the tangent line to the graph of $y = f^{-1}(x)$ at $x = 5$.

1. Find a such that $f(a) = 5$: By inspection, $f(1) = 1^5 + 3(1)^3 + 2(1) - 1 = 1 + 3 + 2 - 1 = 5$. So $a = 1$, meaning $f^{-1}(5) = 1$. The point on f^{-1} is $(5, 1)$. 2. Compute $f'(x)$: $f'(x) = 5x^4 + 9x^2 + 2$. At $a = 1$: $f'(1) = 5(1) + 9(1) + 2 = 16$. 3. Slope of f^{-1}: By the inverse derivative formula: $m = (f^{-1})'(5) = \frac{1}{f'(1)} = \frac{1}{16}$. 4. Tangent line: $y - 1 = \frac{1}{16}(x - 5) \Rightarrow \mathbf{y = \frac{1}{16}x + \frac{11}{16}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic antiderivatives

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m+n$. It typically converges more:

a. slowly than Taylor

b. only at 0

c. **rapidly than Taylor for poles** ✓

d. never

Padé: better than Taylor near poles.

22 [Foundational] • Target: Using initial conditions

Water drains from a cone-shaped tank (radius 3, height 6) at $\frac{dV}{dt} = 2$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{3}{6}h$, find $\frac{dh}{dt}$ when $h = 2$.

a. $\frac{2}{\pi}$ ✓

b. $\frac{1}{\pi}$

c. $-\frac{2}{\pi}$

d. $\frac{4}{\pi}$

$V = \frac{\pi}{3}(\frac{3}{6})^2 h^3 \Rightarrow \frac{dV}{dt} = \pi(\frac{3}{6})^2 h^2 \frac{dh}{dt}$. Solving at $h = 2$: $\frac{dh}{dt} = \frac{2}{\pi}$ m/min.

23 [Foundational] • Target: More antiderivatives

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 t e^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$:
$y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 te^t = (2 + c_2)e^t + (2 + c_2)te^t$.
$y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

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24 [Foundational] • Target: Motion applications

$\int \frac{1}{x^2-1}\,dx$ using partial fractions:

a. $\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$

b. $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

c. $-\frac{1}{2}\ln|x^2-1| + C$

d. A and B are equivalent ✓

$\frac{1}{x^2-1} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$. Integrate: $\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C = \frac{1}{2}\ln|\frac{x-1}{x+1}| + C$.

25 [Foundational] • Target: Antiderivatives of composite expressions

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2 h$

b. $\frac{1}{2}\pi r^2 h$

c. $\pi r^2 h$

d. $\frac{1}{3}\pi r^2 h$ ✓

Applying Applications of Integration principles, $V = \frac{1}{3}\pi r^2 h$. Therefore, the verified correct option is $\frac{1}{3}\pi r^2 h$.

26 [Foundational] • Target: Verifying antiderivatives by differentiation

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty} \binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$ ✓

d. $1 - x/2$

Applying Convergence Tests and Power Series principles, $1 - x/2 - x^2/8 - \cdots$ Therefore, the verified correct option is $1 - x/2 - x^2/8$.

27 [Foundational] • Target: Antiderivatives of reciprocal and rational-power functions

The antiderivative of $f(x) = 0$ is:

a. 0

b. x

c. **any constant C** ✓

d. undefined

Applying Integrals principles, $\int 0\,dx = C$ (any constant). Therefore, the verified correct option is any constant C.

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28 [Foundational] • Target: A projectile motion application

Area of triangle with vertices at origin, $(3, 0)$, $(0, 4)$ using integration:

a. 3

b. 12

c. 6 ✓

d. π

Applying Applications of Integration principles, $A = \frac{1}{2}(3)(4) = 6$. Therefore, the verified correct option is 6.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Visualizing an antiderivative family

Evaluate $\iint_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dA$: using Gaussian integral result $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$:

a. π

b. 1 ✓

c. π^2

d. $1/\pi$

Applying Multiple Integrals principles, Separable: $I = 1 \cdot 1 = 1$. Therefore, the verified correct option is 1.

30 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π ✓

d. $\sqrt{\pi}$

$V = 2\pi \int_0^{\sqrt{\pi/2}} x\cos(x^2) dx = \pi[\sin(x^2)]_0^{\sqrt{\pi/2}} = \pi\sin(\pi/2) = \pi.$

31 [Intermediate] • Target: Basic antiderivatives

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$ ✓

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

$S = 2\pi \int_{-R}^{R} \sqrt{R^2 - x^2} \sqrt{1 + x^2/(R^2 - x^2)}\, dx = 4\pi R^2.$

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32 [Intermediate] • Target: Using initial conditions

Power series integration: $\int_0^x \frac{1}{1+t^2}dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} =$

a. $\arctan x$ ✓

b. $\ln(1+x)$

c. $\sin x$

d. $x/(1+x^2)$

Term-by-term integration of geometric series $1/(1+t^2)$.

33 [Intermediate] • Target: More antiderivatives

Using linearisation to approximate $\sqrt{4.02}$:

a. 2.005 ✓

b. 2.01

c. 2.001

d. 2.002

$L(x) = 2 + \frac{1}{4}(x-4)|_{x=4.02} = 2 + 0.005 = 2.005$.

34 [Intermediate] • Target: Motion applications

Volume of a sphere radius R using disc method:

a. $\frac{4}{3}\pi R^3$ ✓

b. $\frac{4}{3}\pi R^2$

c. $4\pi R^2$

d. $2\pi R^3$

$V = \pi\int_{-R}^{R}(R^2 - x^2)dx = \frac{4}{3}\pi R^3$.

35 [Intermediate] • Target: Antiderivatives of composite expressions

Power series represent real analytic functions. Is $f(x) = e^{-1/x^2}$ ($xe0$), $f(0) = 0$ analytic at 0?

a. **No — all derivatives are 0 but $fe0$ near 0** ✓

b. Yes

c. Yes, since it is smooth

d. No, it is not smooth

Famous example: C^∞ but not analytic at 0.

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36 [Intermediate] • Target: Verifying antiderivatives by differentiation

Shooting angle for max range of a projectile (no air resistance):

a. $30°$

b. $45°$ ✓

c. $60°$

d. $90°$

$R = v^2 \sin 2\theta / g$; max when $\theta = 45°$.

37 [Intermediate] • Target: Antiderivatives of reciprocal and rational-power functions

If $f(0) = 1$, $f'(x) = f(x)^2$, and f has a power series, what is $f(x)$?

a. e^x

b. $\frac{1}{1-x}$ ✓

c. $\ln(1+x)$

d. $\cos x$

ODE $f' = f^2$, $f(0) = 1 \Rightarrow f = 1/(1-x)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A projectile motion application • Cross-Lesson: 15.01 Double Integrals

Maclaurin series for e^x: first four terms:

a. $1 + x + x^2 + x^3$

b. $1 + x + x^2/2! + x^3/3!$ ✓

c. $x + x^2 + x^3$

d. $1 - x + x^2 - x^3$

$e^x = \sum x^n/n! = 1 + x + x^2/2 + x^3/6 + \cdots$

39 [Challenging] • Target: Visualizing an antiderivative family • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Power series of $\arcsin x$: first two non-trivial terms beyond x:

a. $x + x^3/3$

b. $x - x^3/6$

c. $x + x^3/6 + \cdots$ ✓

d. $\pi/2 - x$

$\arcsin x = x + x^3/6 + 3x^5/40 + \cdots$

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40 [Challenging] • Target: Antiderivatives requiring algebraic rewriting first • Cross-Lesson: 2.01 Evaluating Limits

The Gaussian integral: $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$. Using the polar trick $I^2 = \iint e^{-(x^2+y^2)}dA = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta$:

a. $I = \pi$

b. $I = \sqrt{\pi}$ ✓

c. $I = \pi/2$

d. $I = 1$

$I^2 = \pi \Rightarrow I = \sqrt{\pi}$.

41 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 2.02 Continuity

Equation of motion in rotating frame: $m\mathbf{a}_{\text{rot}} = \mathbf{F} - 2m\boldsymbol{\omega} \times \mathbf{v} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. The term $-2m\boldsymbol{\omega} \times \mathbf{v}$ is the:

a. centrifugal force

b. **Coriolis force** ✓

c. gravity

d. normal force

Coriolis force in rotating frame.

42 [Challenging] • Target: Using initial conditions • Cross-Lesson: 3.01 The Derivative from First Principles

A water tank (cylinder, radius 1 m) drains. If height decreases at 0.5 m/min, water volume leaving per minute:

a. 0.5π ✓

b. π

c. 2π

d. 4π

$V = \pi r^2 h \Rightarrow dV/dt = \pi(1)^2(-0.5) = -0.5\pi$ m³/min.

43 [Challenging] • Target: More antiderivatives • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Which of the following series converges?

a. $\sum_{n=2}^{\infty} \frac{1}{2n+1}$

b. $\sum_{n=2}^{\infty} \frac{1}{n\ln(n)^2}$ ✓

c. $\sum_{n=2}^{\infty} \left(1+\frac{1}{n}\right)^n$

d. $\sum_{n=2}^{\infty} \frac{n}{n^2+1}$

$\sum_{n=2}^{\infty} \frac{1}{n\ln(n)^2}$ converges by integral test. The others diverge: $\sum_{n=2}^{\infty} \frac{1}{2n+1}$ by limit comparison with 1/n; $\sum_{n=2}^{\infty} \left(1+\frac{1}{n}\right)^n$ by nth-term test (terms $\to$ e); $\sum_{n=2}^{\infty} \frac{n}{n^2+1}$ by limit comparison with 1/n.

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44 [Challenging] • Target: Motion applications • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Which of the following series diverges?

a. $\sum_{n=2}^{\infty} \left(\frac{n}{2n+1}\right)^n$

b. $\sum_{n=2}^{\infty} \frac{1}{n^2+1}$

c. $\sum_{n=2}^{\infty} \frac{1}{2n+1}$ ✓

d. $\sum_{n=2}^{\infty} n^{-n}n!$

$\sum_{n=2}^{\infty} \frac{1}{2n+1}$ diverges by limit comparison with 1/n. The others converge: $\sum_{n=2}^{\infty} \left(\frac{n}{2n+1}\right)^n$ by root test (limit 1/2); $\sum_{n=2}^{\infty} \frac{1}{n^2+1}$ by comparison with p-series (p = 2); $\sum_{n=2}^{\infty} n^{-n}n!$ by ratio test (limit 1/e < 1).

45 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$?

a. **9** ✓

b. 12

c. 27

d. 3

The integral of $f(x) = x^2$ is $\frac{x^3}{3}$. Evaluating this from 0 to 3 gives: $\left[\frac{x^3}{3}\right]_0^3 = \frac{3^3}{3} - \frac{0^3}{3} = \frac{27}{3} - 0 = 9$. Therefore, the value of the definite integral is 9.

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